

The subcaption package*

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Abstract

This package supports typesetting of sub-captions (by using the the sub-caption feature of the caption package).

- ▲ At the end of each section, text marked with the mountain symbol will contain background knowledge on how the particular command or environment is actually implemented. If you just want to use this package as it is, you don't have to read or understand them.
- ▲ This package demonstrates the usage of `\DeclareCaptionSubType`, `\captionsetup{sub-type}`, and the internal hook `\caption@subtypehook` (offered by the caption package).



Please note: This package is incompatible with the subfigure and subfig packages.

*This package has version number v1.5.

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1 Loading the package

Load this package using

```
\usepackage[\langle options \rangle]{subcaption} .
```

The options for the subcaption package are the same ones as for the caption package, but specify settings which are used for sub-captions *additionally*. In fact

```
\usepackage[\langle options \rangle]{subcaption}
```

is identical to

```
\usepackage{subcaption}  
\captionsetup[sub]{\langle options \rangle} .
```

The default settings for subcaptions are:

```
margin=0pt, font+=smaller, labelformat=parens, labelsep=space,  
skip=6pt, list=false, hypcap=false1
```

Options specified with `\usepackage[...]{subcaption}` and `\captionsetup[sub]{...}` will override the ones specified by `\captionsetup{...}` and `\captionsetup[figure]{...}`, but are again overwritten by `\captionsetup[subfigure]{...}` (same for ‘table’). So finally we have the following order how settings for sub-captions are applied:

1. Global settings (`\usepackage[...]{caption}` and `\captionsetup{...}`)
2. Environmental settings (`\captionsetup[figure -or- table]{...}`)
3. Local settings (`\captionsetup{...}` inside figure or table environment)
4. Default ‘sub’ settings (`margin=0pt, font+=smaller, ...`, see above)
5. Custom ‘sub’ settings (`\usepackage[...]{subcaption}` and `\captionsetup[sub]{...}`)
6. Environmental ‘sub’ settings (`\captionsetup[subfigure -or- subtable]{...}`)
7. Local ‘sub’ settings (`\captionsetup{...}` inside subfigure or subtable)

An example:

```
\usepackage[labelsep=quad, indentation=10pt]{caption}  
\usepackage[labelfont=bf, list=true]{subcaption}  
\captionsetup[table]{textfont=it, position=top}  
\captionsetup[subtable]{textfont=sf}
```

causes the captions inside subtable environments to be typeset with the settings

```
indentation=10pt, position=top, margin=0pt, font=small,  
labelformat=parens, labelsep=space, skip=6pt, hypcap=false,  
labelfont=bf, list=true, textfont=sf .
```

¹This means that sub-captions are not listed in the List of Figures or Tables by default, but you can enable that by specifying the option `list=true`.

2 The `\subcaptionbox` command

`\subcaptionbox` The easiest way to apply a sub-caption is by using the `\subcaptionbox` command, which automatically aligns the sub-figures resp. sub-tables by their very first caption line. Its syntax is:

```
\subcaptionbox[⟨list entry⟩]{⟨heading⟩}[⟨width⟩][⟨inner-pos⟩]{⟨contents⟩}
\subcaptionbox*{⟨heading⟩}[⟨width⟩][⟨inner-pos⟩]{⟨contents⟩}
```

The arguments `⟨list entry⟩` & `⟨heading⟩` will be used for typesetting the `\caption`.

`⟨width⟩` is the width of the resulting `\parbox`; the default value is the width of the contents.

`⟨inner-pos⟩` specifies how the contents will be justified inside the resulting `\parbox`; it can be either ‘c’ (for `\centering`), ‘l’ (for `\raggedright`), ‘r’ (for `\raggedleft`), or ‘s’ (for no special justification). The default is ‘c’. (But you can use any justification defined with `\DeclareCaptionJustification` as well, e.g. ‘centerlast’.)

Using `\subcaptionbox`, the baseline of the resulting box will be placed right between contents and heading, so usually you don’t have to care about the vertical alignment of the sub-figures for yourself. Also the hyperlink anchor is placed properly with respect to the `hypcap=` setting.

One example:

```
\begin{figure}
\centering
\subcaptionbox{A cat\label{cat}}
{\includegraphics{cat}}
\subcaptionbox{An elephant\label{elephant}}
{\includegraphics{elephant}}
\caption{Two animals}\label{animals}
\end{figure}
```

gives the result:

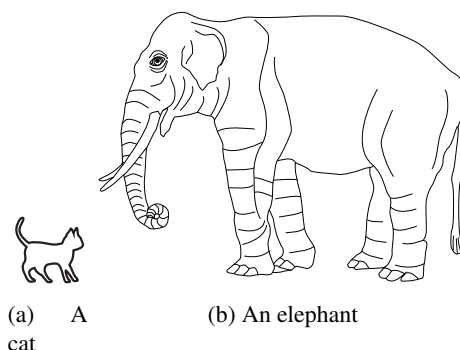


Figure 1: Two animals²

²The pictures were taken with permission from the L^AT_EX Companion[1] examples.

As you see the result is not satisfying; the caption below the cat looks ugly because of the small width of the graphic. This can be solved by using the optional arguments of `\subcaptionbox`, increasing the width of the resulting box:

```
...
\subcaptionbox{A cat\label{cat}}
[2.5cm]{\includegraphics{cat}}
...
```

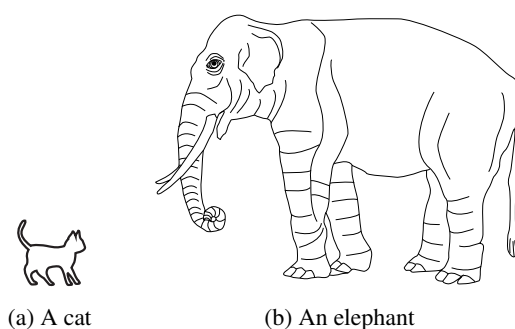


Figure 2: Two animals

Furthermore the main caption, which is centered with respect to the `\textwidth`, looks mis-aligned with respect to the sub-captions. This can (again) be solved by using the optional arguments of `\subcaptionbox`, giving both boxes the same width, for example:

```
...
\subcaptionbox{A cat\label{cat}}
[.4\linewidth]{\includegraphics{cat}}%
\subcaptionbox{An elephant\label{elephant}}
[.4\linewidth]{\includegraphics{elephant}}
...
```

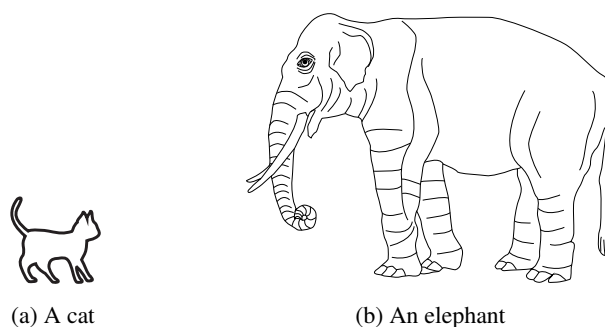


Figure 3: Two animals

▲ The `\subcaptionbox` is a `\parbox` with `\setcaptionsubtype` as first contents line.

2.1 Comparison with `\captionbox`

Both, `\captionbox` (offered by the `caption` package) and `\subcaptionbox`, put its contents and caption into a `\parbox` of either natural or given width and share the same (mandatory and optional) arguments, but while `\captionbox` uses a regular caption, `\subcaptionbox` uses a sub-caption instead, like “(a)” or “2.1”.

So for example the last example would look like this when using `\captionbox` instead of `\subcaptionbox`:

```
\begin{figure}
  \centering
  \captionbox{A cat\label{cat}}
    [.4\linewidth]{\includegraphics{cat}}%
  \captionbox{An elephant\label{elephant}}
    [.4\linewidth]{\includegraphics{elephant}}
\end{figure}
```



Figure 4: A cat

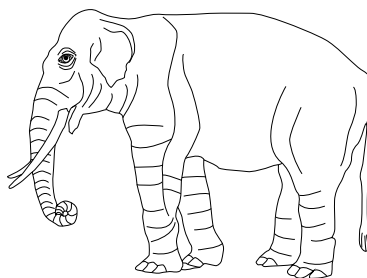


Figure 5: An elephant

3 The subfigure & subtable environments

`subfigure` After loading the `subcaption` package the new environments `subfigure` and `subtable` are available, which have the same (optional & mandatory) arguments as the `minipage` environment:

```
\begin{subfigure} [<outer-pos>] [<height>] [<inner-pos>] {<width>}
...
\end{subfigure}
```

and

```
\begin{subtable} [<outer-pos>] [<height>] [<inner-pos>] {<width>}
...
\end{subtable}
```

Inside these environments you use the ordinary `\caption` command for typesetting captions. So this example is the same as the last one, but uses the `subfigure` environment:

```

\begin{figure}
\begin{subfigure}[b]{.5\linewidth}
\centering\large A
\caption{A subfigure}\label{fig:1a}
\end{subfigure}%
\begin{subfigure}[b]{.5\linewidth}
\centering\large B
\caption{Another subfigure}\label{fig:1b}
\end{subfigure}
\caption{A figure}\label{fig:1}
\end{figure}

```

Some additional notes:

- You can override the settings for a specific subcaption with a `\captionsetup` inside the subfigure or subtable environment, e.g.:

```

\begin{subfigure}[b]{.5\linewidth}
\centering\large A
\captionsetup{skip=3pt}
\caption{A subfigure}\label{fig:1a}
\end{subfigure}

```

- A subfigure environment could have multiple captions, e.g.:

```

\begin{subfigure}[b]{.5\linewidth}
\centering\large A
\caption{Subfigure A}\label{subfig:A}
\medskip
\centering\large B
\caption{Subfigure B}\label{subfig:B}
\end{subfigure}

```

- The environment name must match the current floating environment, i.e. inside a figure a subfigure should be used, and inside a table a subtable should be used. Using the wrong sub-type is not supported and causes an error since v1.4 of the subcaption package.
- Hyperlinks targeted to this subfigure will jump to the beginning of the subfigure, and not to the caption of the subfigure (if `hypcap=true` is set for sub-captions). (See [section 6.6: Where do hyperlinks jump?](#))

New feature
v1.2

Beside the *outer-pos* values of ‘c’, ‘t’, and ‘b’, the subcaption package also offers the values ‘T’ and ‘B’ additionally which align the subfigure at the very top resp. bottom. (In contrast ‘t’ and ‘b’ align the subfigure at the top resp. bottom *baseline*.)



The subfigure & subtable environments are just simple minipage environments with `\setcaptionsubtype` as first contents line. These environments are defined with the help of `\ForEachCaptionSubType`, which executes code for every sub-type declared with `\DeclareCaptionSubType`.

4 The `\subcaption` command

`\subcaption` The most flexible way to apply a sub-caption is by using the `\subcaption` command. Its syntax is analogous to the one of the `\caption` command and shares its features:

```
\subcaption[<list entry>]{<heading>}
\subcaption*{<heading>}
```

Please note that the `\subcaption` command *must* be applied inside its own box or environment.

An example:

```
\begin{figure}
\begin{minipage}[b]{.5\linewidth}
\centering\large A
\subcaption{A subfigure}\label{fig:1a}
\end{minipage}%
\begin{minipage}[b]{.5\linewidth}
\centering\large B
\subcaption{Another subfigure}\label{fig:1b}
\end{minipage}
\caption{A figure}\label{fig:1}
\end{figure}
```

gives the result:

A (a) A subfigure	B (b) Another subfigure
--	--

Figure 6: A figure

▲ The `\subcaption` command is just a simple combination of `\setcaptionsubtype*` and `\caption`.

5 The `\DeclareCaptionSubType` command

`\DeclareCaptionSubType` For using the sub-caption feature of the caption package some commands and counters must be prepared. This is done with

```
\DeclareCaptionSubType[<numbering scheme>]{<type>}
\DeclareCaptionSubType*[<numbering scheme>]{<type>}
```

For the environments `figure` & `table`, and all the ones defined with `\DeclareFloatingEnvironment` offered by the `newfloat` package, this will be done automatically, but for other environments (e.g. the ones defined with `\newfloat` offered by the `float` package or `\DeclareNewFloatType` offered by the `floatrow` package) this has to be done manually.

The starred variant provides the sub-caption numbering format `<type>.<subtype>` (for example ‘1.2’) while the non-starred variant simply uses `<subtype>` (for example ‘a’).

Own numbering formats can be created by redefining `\thesub<type>`, e.g.:

```
\DeclareCaptionSubType*{figure}
\renewcommand\thesubfigure{\thefigure\alph{subfigure}}
```

would give you sub-caption numbers like ‘1b’.

The default numbering scheme is `alph`, but you can use any $\LaTeX$ (or self-defined) command name here which converts a counter to a text value, e.g.: `arabic`, `roman`, `Roman`, `alph`, `Alph`, `fnsymbol`, ...

But `\DeclareCaptionSubType` is not only for defining new sub-caption types, you can use this command for re-definitions as well, e.g.:

```
\DeclareCaptionSubType*[arabic]{table}
\captionsetup[subtable]{labelformat=simple,labelsep=colon}
```

will give you sub-captions in tables like these ones:

Table 1: Two tables			
1.1: Table one		1.2: Table two	
A	B	E	F
C	D	G	H

▲ `\DeclareCaptionSubType` and `\ForEachCaptionSubType` are integral parts of the caption package kernel.

6 References

The macro `\the<counter>` is not only responsible for the look of the `<counter>`, but for the look of the references typeset with `\ref`, too. References will be prefixed by $\LaTeX$ with the internal macro `\p@<counter>`.

`\DeclareCaptionSubType` will define both of them for sub-captions (e.g. `subfigure` and `subtable`), and as you have seen in the last section `\DeclareCaptionSubType` will give you some options to control the internal (re-)definition of `\the<counter>` and `\p@<counter>`.

`\thesubfigure` For example `\thesubfigure` and `\p@subfigure` are (as default) internally defined as

```
\newcommand\thesubfigure{\alph{subfigure}}
\newcommand\p@subfigure{\thefigure}
```

so the label of sub-captions will look like ‘a’ (decorated by the selected label format), while references will look like ‘1a’ since they are prefixed by `\p@subfigure = \thefigure`.

After `\DeclareCaptionSubType*[arabic]{figure}`, `\thesubfigure` and `\p@subfigure` will look like

```
\renewcommand\thesubfigure{\thefigure.\arabic{subfigure}}
\renewcommand\p@subfigure{}
```

But if you want detailed control on how the references will look like, the options of `\DeclareCaptionSubType` are potentially not sufficient. In this case one need to redefine these two macros on his/her own. Some examples:

If you want parentheses around the sub-figure part of the reference, so they will look like ‘1 (a)’, you may get them this way:

```
\usepackage[labelformat=simple]{subcaption}
\renewcommand\thesubfigure{(\alph{subfigure})}
```

(Note: Since `parens` is the default label format you will get double parentheses in sub-captions when not specifying a different label format, e.g. `simple`.)

But if you want only a closing parenthesis, so references will look like ‘1a)’, but the sub-captions itself should still look like ‘(a)’, this would be a possible solution:

```
\usepackage{subcaption}
\renewcommand\thesubfigure{\alph{subfigure}}
\DeclareCaptionLabelFormat{opening}{(#2)}
\captionsetup[subfigure]{labelformat=opening}
```

(Please note that you need to surround redefinitions of `\p@<counter>` with `\makeatletter` and `\makeatother`. See <http://tex.stackexchange.com/questions/8351/> for details.)

6.1 The `\subref` command

While `\ref{<key>}` (and `\ref*{<key>}`, if the `hyperref` package is used) usually gives a combined result representing the main caption counter and the sub-caption one, it is sometimes useful to have a reference to the sub-caption only. For this purpose you can use

```
\subref{<key>}
\subref*{<key>}3
```

So for example `\ref{cat}` gives the result ‘1a’ but `\subref{cat}` gives ‘a’.

Note: If the sub-caption was (re-)defined with the starred variant `\DeclareCaptionSubType*`, both `\ref` and `\subref` usually gives the same result.

▲ The `\subref` command demonstrates the usage of `\caption@subtypehook` which will be called during `\captionsetup{subtype}`.

6.2 The `subrefformat=` option

subrefformat= By applying `\DeclareCaptionSubType`, or by redefining `\the<counter>` and `\p@<counter>`, you will change the look of references typeset with `\ref` and `\subref`. But maybe you only want to change the output of `\subref` without affecting the references typeset with `\ref`? This is possible, too, by using the option `subrefformat`:

New feature
v1.1

```
\captionsetup[subrefformat=<label format>]
```

³Like `\ref*`, `\subref*` is only available if the `hyperref` package^[3] is used.

This one will choose a label format (either a pre-defined one, or a one defined with `\DeclareCaptionLabelFormat`) as decorative element to sub-references. The default one is `simple` which has no decorative elements but simply typeset the reference as it is.

For example

```
\captionsetup{subrefformat=parens}
```

will result in references (typeset with `\ref`) like ‘1a’ but sub-references (typeset with `\subref`) like ‘(a)’.

6.3 Referencing sub-figures without sub-captions

`\phantomsubcaption`
`\phantomcaption`

New feature
v1.1

If you don’t want to give a sub-figure a caption (yet), because the picture itself already contains the caption, or for some other reason, you could use the command

```
\phantomsubcaption
```

instead of `\subcaption`, or – when inside a `subfigure` or `subtable` environment – `\phantomcaption` instead of `\caption`. `\phantomsubcaption` and `\phantomcaption` do not have any arguments, and they do not generate any output, but increase the sub-figure resp. sub-table counter and give you an anchor for a `\label` command which can be placed afterwards.

Please note that – just like `\subcaption` – the `\phantomsubcaption` command *must* be applied inside its own group, box, or environment.

An example:

```
\begin{figure}
\centering
{\includegraphics{cat_with_a}
\phantomsubcaption\label{cat}}
{\includegraphics{elephant_with_b}
\phantomsubcaption\label{elephant}}
\caption{Two animals: \subref{cat} a huge cat,
and \subref{elephant} an elephant}
\end{figure}
```

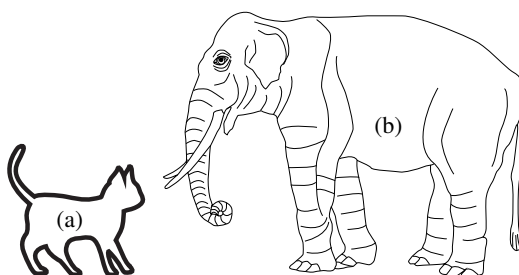


Figure 7: Two animals: (a) a huge cat, and (b) an elephant

- ▲ The `\phantomsubcaption` command is just a simple combination of `\setcaptionsubtype*` and `\phantomcaption`.

`\subcaptionlistentry` If you don't want to give a sub-figure a caption (yet), because the picture itself already contains the caption, or for some other reason, you could also use the command `\captionlistentry`

New feature
v1.5

`\subcaptionlistentry{⟨list entry⟩}`

instead of `\subcaption`, or – when inside a subfigure or subtable environment – `\captionlistentry` instead of `\caption`. `\subcaptionlistentry` and `\captionlistentry` do not generate any output but increase the sub-figure resp. sub-table counter and give you an anchor for a `\label` command which can be placed afterwards. Additionally to `\phantomsubcaption` and `\phantomcaption` these commands put an entry into the list of figures resp. tables.

Please note that – just like `\subcaption` – the `\subcaptionlistentry` command *must* be applied inside its own group, box, or environment.

- ▲ The `\subcaptionlistentry` command is just a simple combination of `\setcaptionsubtype*` and `\captionlistentry`.

6.4 Typesetting sub-captions without generating a (new) reference

`\subcaptiontext`
`\captiontext`

New feature
v1.5

The `\caption` resp. `\subcaption` command is a multi-purpose command:

1. It does `\setcaptionsubtype*` (`\subcaption` only)
2. It increments the sub-figure resp. sub-table counter and generates an internal reference which could be used with `\label`
3. It puts an entry into the list of figures resp. tables
4. It finally typesets a caption

When put into a command or into an environment which either evaluates its content more than once or does not like one of the first three actions (for whatever reason), the result could be either an error message or an incorrect result, for example a figure resp. table counter which was incremented more than once. In this cases the `\caption` resp. `\subcaption` command could be split into `\phantomcaption` which performs step two only (or `\captionlistentry` which performs steps two and three) resp. `\phantomsubcaption` which performs steps two and three (or `\subcaptionlistentry` which performs steps one to three), and `\captiontext` which performs step four only (or `\subcaptiontext` which performs the steps one and four). This way critical steps could be out-sourced from the target command or environment, for example by prepending `\phantomsubcaption` and using `\captiontext` inside.

The syntax of `\subcaptiontext` is

```
\subcaptiontext[⟨number⟩]{⟨text of sub-caption⟩}
\subcaptiontext*[⟨number⟩]{⟨text of sub-caption⟩}
```

and it behaves like `\captiontext` offered by the `caption` package but typesets a subcaption instead of a caption.

Please note that – just like `\subcaption` – the `\subcaptiontext` command *must* be applied inside its own group, box, or environment.

For example code please take a look at [section 7: Captions inside sub-figures](#).

▲ The `\subcaptiontext` command is just a simple combination of `\setcaptionsubtype*` and `\captiontext`.

6.5 Where to place the `\label` command?

When using `\subcaption`, or `\caption` inside a subfigure or subtable environment, the `\label` can be either placed inside the caption text or right after the command, e.g.:

```
\subcaption{Some text here\label{label1}}
...
\subcaption{Some other text}\label{label2}}
...
\subcaption{Something different}
\label{label3}
```

When using `\phantomsubcaption` or `\subcaptionlistentry`, or `\phantomcaption` or `\captionlistentry` inside a subfigure or subtable environment, the `\label` should be placed right after the command, e.g.:

```
\phantomsubcaption\label{label4}}
...
\phantomsubcaption
\label{label5}
```

But when using the `\subcaptionbox` command, the `\label` should be placed inside the caption text, e.g.:

```
\subcaptionbox{A description here\label{label6}}
{Some content here}
...
\subcaptionbox[List-of-Figures entry]
{A description here\label{label7}}
{Some content here}
```

Placing `\label` outside the `\subcaptionbox` would produce an incorrect reference.

6.6 Where do hyperlinks jump?

For the subfigure & subtable environments and `\subcaptionbox` boxes (and own constructs which use `\setcaptionsubtype`) the hyperlink anchors will be placed in respect to the `hycap=` setting. While usage of this option is straight-forward

for ordinary captions, the usage for sub-captions depends on the setting regarding the main captions. This table gives you an overview where the hyperlinks will jump:

caption subcaption	hypcap=false	hypcap=true
	sub-caption	figure or table (default setting)
hypcap=false		
hypcap=true	sub-figure or sub-table	sub-figure or sub-table

But if `\subcaption` is used and `hypcap=true` is set for sub-captions, the `subcaption` package does not know where the sub-figure or sub-table actually begins, so it will jump to the sub-caption instead.

Remember: If you use the `hypcap` package^[4], it controls the placement of the hyperlink anchors, making the rules above invalid.

(See also the documentation of the `caption` package, sections about `hyperref` & `hypcap`.)

7 Captions inside sub-figures

Nearly all code examples so far have placed the caption either above or below the sub-figure. But it is possible to put the caption inside the sub-figure, too:

- The caption could already be part of the image. (This case is already handled in [section 6.3: Referencing sub-figures without sub-captions.](#))
- $\text{\LaTeX}$ packages like `overpic`, `stackengine`, or `tikz` could be used. Note that the commands resp. environments offered by these packages usually evaluate their content more than once, resulting in either errors or wrong reference counters. For this reason `\subcaption` should not be used here, instead `\subcaptionlistentry` resp. `\phantomsubcaption` should be used outside the command resp. environment and `\subcaptiontext` should be used inside it. See also [section 6.4: Typesetting sub-captions without generating a \(new\) reference](#)
- The options `skip=` and `margin=` could be used to place the caption onto the image.

7.1 Using the `overpic` package

The same example as in [section 6.3: Referencing sub-figures without sub-captions](#), but using the `overpic` environment offered by the `overpic` package to place the captions inside the pictures:

```

\usepackage{overpic}
...
\begin{figure}
  \centering
  {\subcaptionlistentry{}\label{cat}
   \begin{overpic}[width=60pt]{cat}
     \put(40,34){\captiontext*{}}
   \end{overpic}}
  {\subcaptionlistentry{}\label{elephant}
   \begin{overpic}[width=140pt]{elephant}
     \put(58,40){\captiontext*{}}
   \end{overpic}}
  \captionsetup{subrefformat=parens}
  \caption{Two animals: \subref{cat} a huge cat,
    and \subref{elephant} an elephant}
\end{figure}

```

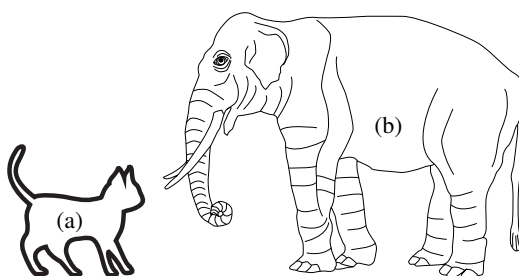


Figure 8: Two animals: (a) a huge cat, and (b) an elephant

If no references to the sub-figures are needed, the usage of an extra group plus `\phantomsubcaption` resp. `\subcaptionlistentry` could be dropped. Since this leaves `\captiontext` without a valid sub-figure number value, it must be given to it as optional argument. Furthermore `\subcaptiontext` must be used here since `\captiontext` would typeset a main figure caption (and not a sub-figure one) since there is no other command left to switch into the sub-figure mode. (In the example code above both could be used, `\captiontext` and `\subcaptiontext`, since `\subcaptionlistentry` has already triggered the sub-figure mode there. When in doubt, don't think about it and just always use `\subcaptiontext` for sub-captions.)

```

\usepackage{overpic}
...
\begin{figure}
  \centering
  \begin{overpic}[width=60pt]{cat}
    \put(40,34){\subcaptiontext*[1]{}}
  \end{overpic}
  \begin{overpic}[width=140pt]{elephant}
    \put(58,40){\subcaptiontext*[2]{}}
  \end{overpic}
\end{figure}

```

```

\captionsetup{subrefformat=parens}
\caption{Two animals: A huge cat and an elephant}
\end{figure}

```

7.2 Using the stackengine package

The same example as in section [section 6.3: Referencing sub-figures without sub-captions](#), but using the `\stackinset` command offered by the `stackengine` package to place the captions inside the pictures:

```

\usepackage{stackengine}
...
\begin{figure}
\centering
{\subcaptionlistentry{\label{cat}
\stackinset{l}{25pt}{b}{20pt}{\subcaptiontext*{}}
{\includegraphics[width=60pt]{cat}}}}
{\subcaptionlistentry{\label{elephant}
\stackinset{l}{80pt}{b}{60pt}{\subcaptiontext*{}}
{\includegraphics[width=140pt]{elephant}}}}
\captionsetup{subrefformat=parens}
\caption{Two animals: \subref{cat} a huge cat,
and \subref{elephant} an elephant}
\end{figure}

```

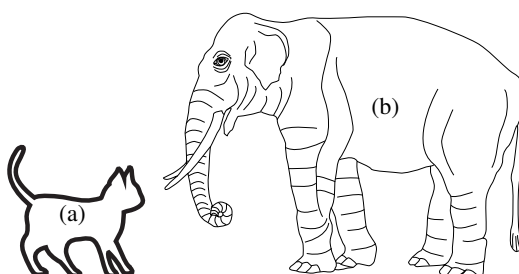


Figure 9: Two animals: (a) a huge cat, and (b) an elephant

If no references to the sub-figures are needed, the usage of an extra group plus `\phantomsubcaption` could be dropped here, too:

```

\usepackage{stackengine}
...
\begin{figure}
\centering
\stackinset{l}{25pt}{b}{20pt}{\subcaptiontext*[1]{}
{\includegraphics[width=60pt]{cat}}}
\stackinset{l}{80pt}{b}{60pt}{\subcaptiontext*[2]{}
{\includegraphics[width=140pt]{elephant}}}

```

```

\captionsetup{subrefformat=parens}
\caption{Two animals: A huge cat and an elephant}
\end{figure}

```

7.3 Using the tikz package

The same example as in section [section 6.3: Referencing sub-figures without sub-captions](#), but using the `tikzpicture` environment offered by the `tikz` package to place the captions inside the pictures:

```

\usepackage{tikz}
...
\begin{figure}
\centering
{\subcaptionlistentry{}\label{cat}
\begin{tikzpicture}
\node (cat) at (0,0)
{\includegraphics[width=60pt]{cat}};
\node at (0.1,-0.1) {\subcaptiontext*{}};
\end{tikzpicture}}
{\subcaptionlistentry{}\label{elephant}
\begin{tikzpicture}
\node (elephant) at (0,0)
{\includegraphics[width=140pt]{elephant}};
\node at (0.5,-0.1) {\subcaptiontext*{}};
\end{tikzpicture}}
\captionsetup{subrefformat=parens}
\caption{Two animals: \subref{cat} a huge cat,
and \subref{elephant} an elephant}
\end{figure}

```

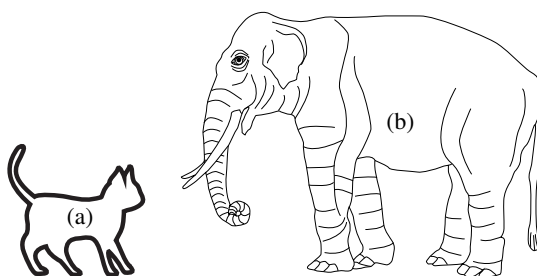


Figure 10: Two animals: (a) a huge cat, and (b) an elephant

If no references to the sub-figures are needed, the usage of an extra group plus `\phantomsubcaption` could be dropped here, too:

```

\usepackage{tikz}
...

```

```

\begin{figure}
\centering
\begin{tikzpicture}
\node (cat) at (0,0)
{\includegraphics[width=60pt]{cat}};
\node at (0.1,-0.1) {\subcaptiontext*[1]{} };
\end{tikzpicture}
\begin{tikzpicture}
\node (elephant) at (0,0)
{\includegraphics[width=140pt]{elephant}};
\node at (0.5,-0.1) {\subcaptiontext*[2]{} };
\end{tikzpicture}
\captionsetup{subrefformat=parens}
\caption{Two animals: \subref{cat} a huge cat,
and \subref{elephant} an elephant}
\end{figure}

```

7.4 Using the skip and margin options

For a particular sub-caption the `skip=...` could be set to a negative value so it will overlap with the image. Combined with `singlelinecheck=off` (to switch off the centering of short captions) and `margin=...` it could be placed at a specific horizontal position within the image, too.

The same example as in [section 6.3: Referencing sub-figures without sub-captions](#), but using the `skip=...` and `margin=...` options:

```

\begin{figure}
\centering
\captionsetup[subfigure]
{skip=-28pt,slc=off,margin={25pt,0pt}}
\subcaptionbox{\label{cat}}
{\includegraphics[width=60pt]{cat}}
\captionsetup[subfigure]
{skip=-60pt,slc=off,margin={80pt,0pt}}
\subcaptionbox{\label{elephant}}
{\includegraphics[width=140pt]{elephant}}
\captionsetup{subrefformat=parens}
\caption{Two animals: \subref{cat} a huge cat,
and \subref{elephant} an elephant}
\end{figure}

```

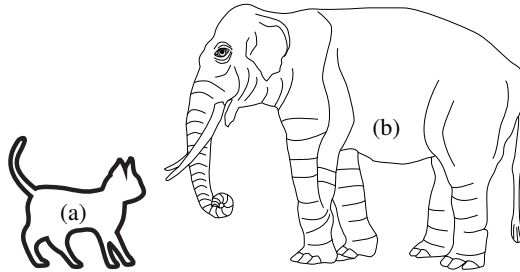


Figure 11: Two animals: (a) a huge cat, and (b) an elephant

8 Numbering

8.1 Pitfall #1: Using multiple main captions

When multiple main captions are used within a figure or table, and sub-captions are used as well, how does the `subcaption` package know which sub-captions belong to which caption, i.e. what is the main counter value for the sub-captions?

Let's illustrate this problem with an example document:

```
\documentclass{article}
\usepackage{graphicx,subcaption}
\begin{document}
\begin{figure}
\centering
\includegraphics[width=4cm]{example-image-c}
\caption{Caption no. 1}
\bigskip
\subcaptionbox
{\label{fig:2a}}
{\includegraphics[width=2cm]{example-image-a}}
\subcaptionbox
{\label{fig:2b}}
{\includegraphics[width=2cm]{example-image-b}}
\caption{Caption no. 2}
\end{figure}
Look at sub-figures \ref{fig:2a} and \ref{fig:2b}.
\end{document}
```

It's obvious that the sub-captions belong to 2nd caption, and therefore `\ref{fig:2a}` will become "2a", isn't it? But since the `subcaption` package is only involved when using $\LaTeX$ commands either defined or patched by the `caption` or `subcaption` package, this is what the `caption` package is aware of:

```
\usepackage{subcaption}
\begin{document}
\begin{figure}
```

```

\caption{Caption no. 1}
\subcaptionbox
  {\label{fig:2a}}
  {\textit{unknown content}}
\subcaptionbox
  {\label{fig:2b}}
  {\textit{unknown content}}
\caption{Caption no. 2}
\end{figure}
\end{document}

```

So from captions point of view it's not easy to decide if the sub-captions belong to the 1st or 2nd main caption since they are placed between them. (Note: The subcaption package is only offering an user interface to the sub-caption feature of the caption package, and therefore this decision is the responsibility of the caption package.)

But how does the caption package makes a decision? If in doubt, it rather clings to the past than to the future, i.e. in this case it decides that the sub-captions belong to the 1st caption and therefore the result of `\ref{fig:2a}` is not “2a” but “1a”. If the caption package is unsure about its decision (like in this case), a warning will be issued:

```

Package caption Warning: Ambiguous sub-caption(s),
                        use \nextfloat on input line 15.
See the caption package documentation for explanation.

```

How to fix it? Use `\nextfloat` to tell the caption package where the 2nd figure within the figure environment starts:

```

\documentclass{article}
\usepackage{graphicx,subcaption}
\begin{document}
\begin{figure}
  \centering
  \includegraphics[width=4cm]{example-image-c}
  \caption{Caption no. 1}
  \bigskip
  \nextfloat
  \subcaptionbox
    {\label{fig:2a}}
    {\includegraphics[width=2cm]{example-image-a}}
  \subcaptionbox
    {\label{fig:2b}}
    {\includegraphics[width=2cm]{example-image-b}}
  \caption{Caption no. 2}
\end{figure}
Look at sub-figures \ref{fig:2a} and \ref{fig:2b}.
\end{document}

```

Note that the `\nextfloat` command was introduced in caption package v3.6. Previous versions of the caption package have bound the decision to the `position=` setting of the floating environment instead, i.e. sub-captions belonged to the caption above if

`position=top` was set and they belonged to the caption below if `position=below` was set. While this would be beneficial in this case, there were several cases where it was not and especially it was not always comprehensible to the user why his references got an incorrect numbering.

Furthermore the new decision algorithm always succeeds if there is only one caption within the figure or table (which is the case most of the time) while the old one did not.

If you still prefer the old decision algorithm (for example because you want to process an already existing document), you need to specify the caption package version explicitly, for example:

```
\documentclass{article}
\usepackage{caption}[v3.5]
\usepackage{graphicx,subcaption}
...
```

See also: [section 10: Required packages](#)

8.2 Pitfall #2: Expecting `\caption` to increment the counter

Usually `\caption` increments the figure resp. table counter and therefore it is usually safe to assume that the counter was not incremented yet in code used before `\caption`:

```
\documentclass{article}
\begin{document}
\begin{figure}
  Figure counter before caption: \thefigure
  \caption{Some text}
  Figure counter after caption: \thefigure
\end{figure}
\end{document}
```

Unsurprisingly the results are 0 and 1 for the counter values. But this happens if we use the `subcaption` package:

```
\documentclass{article}
\usepackage{subcaption}
\begin{document}
\begin{figure}
  \subcaptionbox{}{some content}
  \subcaptionbox{}{some content}
  Figure counter before caption: \thefigure
  \caption{Some text}
  Figure counter after caption: \thefigure
\end{figure}
\end{document}
```

Here the results are 1 and 1 for the counter values. Why?

Since the `caption` package assumes that there will be a `\caption` following `\subcaptionbox` it decides that the sub-captions belong to the upcoming main caption and

therefore share the same main counter value which needs to be incremented before its use. Therefore the first `\subcaptionbox` in the figure increments the figure counter while the second one and especially the `\caption` does not.

Usually this is no problem at all but keep this in mind if you are doing tricky stuff with the figure resp. table counter within figures resp. tables.

9 The `\subfloat` command

`\subfloat` To allow a smoother transition from the `subfig` package (which is unmaintained for over 15 years) this package also offers `\subfloat` with the same syntax:

New feature v1.3

```
\subfloat[⟨list_entry⟩][⟨sub-caption⟩]{⟨body⟩}
```

10 Required packages

Starting with version 1.4 the `subcaption` package requires at least version 3.1 of the `caption` package and loads it automatically. (Older versions of the `subcaption` package have required exactly the version of the `caption` package which was released with it.)

New feature v1.4

If you need to use a specific version of the `caption` package you need to load it *before* the `subcaption` package, e.g.:

```
\usepackage[...]{caption}[=v3.5]
\usepackage[...]{subcaption}
```

Note that there are limitations if an older version of the `caption` package is used:

- The `\phantomsubcaption` command need at least `caption` v3.2.
- The `\subcaptionlistentry` command need at least `caption` v3.3.
- The `\subcaptiontext` command need at least `caption` v3.6.
- The `\subfloat` emulation needs at least `caption` v3.4.

11 Beyond this package

For a more advanced usage of the sub-caption feature of the `caption` package, please take a look at the excellent `keyfloat` package^[5] which provides the environments `keysubfigs`, `keysubtabs`, and `keysubfloats` for typesetting sub-figures and sub-tables. Furthermore the `floatrow` package^[2] provides the `subfloatrow` environment for typesetting sub-figures.

12 Thanks

I would like to thank Stephen Dalton who helped to make this package a better one.

References

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